

Announcements

Topic 3: Graphs

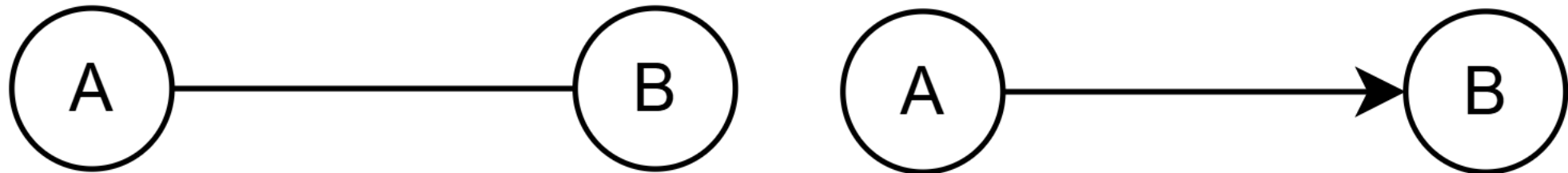
By Professor Hudson Lynam

Graph Definition

- Definition: *Graph*
- *A graph $G=(V,E)$ where V is a finite non-empty set of vertices (a.k.a. nodes) and E is a binary relation on V .*
- Note that definitions of “graph” vary: this is a common definition we’ll be using for this class

Undirected versus Directed Graphs

- Undirected:
 - The pairs of vertices in E are **unordered**. In other words, each edge is a set, like $\{A,B\}$.
 - When we say “ G is a graph,” the default assumption is that G is undirected
- Directed
 - Each edge has an order, like a tuple (A,B) .
 - Because directed graphs are common, we usually abbreviate to “digraph.”



Graphs Example

- What would V be for this graph?
- $V = \{A, B, C, D\}$
- $E = \{\{A, C\}, \{A, B\}, \{A, D\}, \{B, D\}\}$
- What about V and E for this digraph?
- $V = \{1, 3, 5, 7\}$
- $E = \{(1, 3), (3, 1), (5, 1), (7, 7), (7, 5)\}$
- Note: Don't use bidirectional arrows for undirected graphs, and always use arrows to show two edges between vertices in a digraph

